Simulation Studies

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Theory often cannot give the bias, variance or coverage of an estimator when the estimator or the data distribution is complicated. A simulation study measures the same properties for any estimator a computer can run. It generates many data sets from a distribution whose parameters are known, applies the estimator to each, and summarizes the results.

This page defines the parts of a simulation study.

The data-generating process

Definition 0.1 (Data-generating process)**.** The **data-generating process (DGP)** of a simulation is the joint probability distribution F_{θ_0} from which it draws its data $X_1, \dots, X_n$, with the number of observations n and the parameter value θ_0 fixed and known:

$$X_1, \dots, X_n \sim F_{\theta_0}.$$

Example 0.1 (A normal data-generating process)**.**
Let $X_1, \dots, X_n$ be IID normal random variables with expectation $\mu = 0$ and variance $\sigma^2 = 4$, with $n = 10$. The distribution of these variables is a data-generating process, and the estimand $\sigma^2 = 4$ is known exactly, so the error of any estimator of σ^2 can be computed.

Replications and simulation studies

Definition 0.1 (Replication)**.** A **replication** of a simulation is one run of its steps: draw a data set $X_1, \dots, X_n^{(m)}$ from the data-generating process, then compute the estimate $T_m \stackrel{\text{def}}{=} \hat{\theta}\big(X_1, \dots, X_n^{(m)}\big)$, where $\hat{\theta}$ is an estimator, for $m = 1, \dots, M$. Different replications draw their data sets independently.

Example 0.1 (A replication of the normal data-generating process)**.**
For the normal data-generating process, one replication draws $n = 10$ values from the normal distribution with expectation 0 and variance 4. It then computes the estimate T_m of the variance σ^2 from those 10 values. A second replication draws 10 new values, independently of the first, and computes T_{m+1}.

Definition 0.2 (Simulation study)**.** A **simulation study** runs M replications of one data-generating process and one estimator, then summarizes the estimates $T_1, \dots, T_M$ to learn the estimator's sampling distribution.

Example 0.2 (A simulation study of a variance estimator)**.**
For the normal data-generating process, a simulation study of one estimator of the variance σ^2 runs M replications of that process and that estimator. The M estimates $T_1, \dots, T_M$ then show how the estimator behaves across repeated data sets.

Simulation estimates

Definition 0.1 (Simulation estimate)**.** A **simulation estimate** of the expectation $\mathbb{E}[T]$ of an estimate T is the average of the replications' estimates $T_1, \dots, T_M$:

$$\bar{T}_M \stackrel{\text{def}}{=} \frac{1}{M} \sum_{m=1}^{M} T_m.$$

Example 0.1 (A simulation estimate from three replications)**.**
Suppose $M = 3$ replications give the estimates $T_1 = 3$, $T_2 = 5$ and $T_3 = 4$. Then the simulation estimate of $\mathbb{E}[T]$ is

$$\bar{T}_3 = \frac{3 + 5 + 4}{3} = 4.$$

Definition 0.2 (Simulation estimate of the bias)**.** Let θ be an estimand, let $\hat{\theta}$ be an estimator, and let $T_1, \ldots, T_M$ be its estimates from M replications. The **simulation estimate of the bias** of $\hat{\theta}$ is the simulation estimate $\bar{T}_M$ of $\mathbb{E}\left[\hat{\theta}\right]$, minus θ:

$$\bar{T}_M - \theta.$$

Example 0.2 (A simulation estimate of the bias from three replications)**.**
Continue the three replications, and let the estimand be $\theta = 5$. Then the simulation estimate of the bias is

$$\bar{T}_3 - \theta = 4 - 5 = -1.$$

Definition 0.3 (Simulation estimate of the mean squared error)**.** Let θ be an estimand, let $\hat{\theta}$ be an estimator, and let $T_1, \ldots, T_M$ be its estimates from M replications. The **simulation estimate of the mean squared error** of $\hat{\theta}$ is the average of the squared differences $(T_m - \theta)^2$:

$$\frac{1}{M} \sum_{m=1}^{M} (T_m - \theta)^2.$$

Example 0.3 (A simulation estimate of the mean squared error from three replications)**.**

Continue the three replications with $\theta = 5$. The squared differences are $(3-5)^2 = 4$, $(5-5)^2 = 0$ and $(4-5)^2 = 1$, so the simulation estimate of the mean squared error is

$$\frac{4+0+1}{3} = \frac{5}{3}.$$

Remark (Simulation estimates in Bayesian inference)

Remark 0.1 (Simulation estimates in Bayesian inference). The Monte Carlo estimate of a posterior expectation is the same average, taken over draws from a posterior distribution instead of over replications of a data-generating process.

References